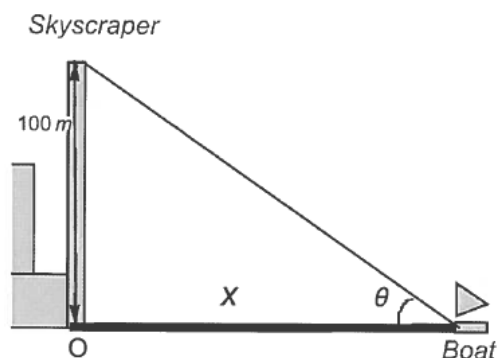


Problem 10

Problem 10

Problem 1 A boat sails directly toward a 100-meter skyscraper that stands on the edge of a harbor. The angular size θ of the building is the angle formed by lines from the top and bottom of the building to the observer on the boat (see figure below).



- (a) Express the angle θ as the function of x , the distance of the boat from the building.
- (b) The boat is sailing directly toward the skyscraper at 3 m/s. Find $\frac{d\theta}{dt}$ when the boat is $x = 300\text{m}$ from the building. [3]